Sensing depth of a periodic electrode array

Why the depth scale is $\lambda/2\pi$, and how much of the signal comes from within depth d

Project Treebeard · MSV4 electrode notes · 30 September 2026

Results. For electrodes that alternate driven and ground with centre-to-centre spacing s, the pattern repeats every $\lambda = 2s$, and

$$\begin{aligned} &\text{1/e depth of the field:} && \delta = \frac{\lambda}{2\pi}, \\ &\text{share from within } d\text{, weak contrast:} && S(d) = 1 - \mathrm{e}^{-4\pi d/\lambda}, \\ &\text{share from within } d\text{, soil } \varepsilon_1 \text{ over air } \varepsilon_2\text{:} && S(d) = \frac{(1+\rho)\tanh(2\pi d/\lambda)}{1+\rho\tanh(2\pi d/\lambda)}, \quad \rho = \frac{\varepsilon_2}{\varepsilon_1}. \end{aligned}$$

The pattern enters only through λ. For a periodic pattern each result is an upper bound on how deep the sensor looks (the third under a fixed surface potential, Section 4); higher harmonics only make the response shallower.

1 Setup

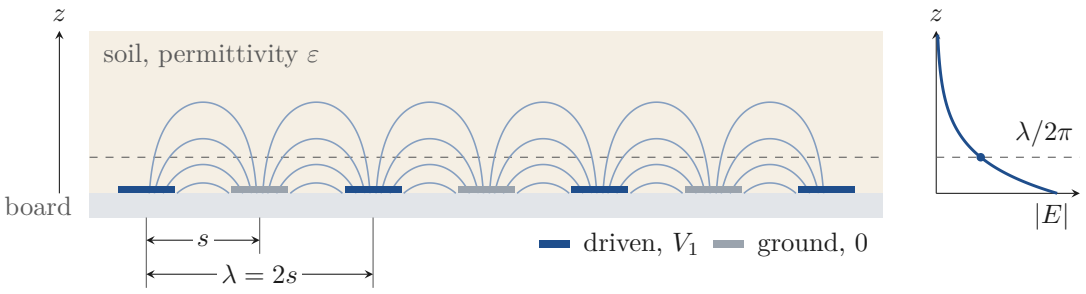


Figure 1: Cross-section across the blade (to scale in units of λ). Electrodes alternate driven and ground at centre spacing s, so the surface pattern repeats every $\lambda = 2s$. The curves are exact field lines of the fundamental harmonic (Section 2). Right: its field magnitude, which has fallen to 1/e at the dashed line $z = \lambda/2\pi$.

Coordinates: x runs across the blade, y along it, and z into the soil from the electrode plane $z = 0$.

(A1) **Two-dimensional, quasi-static.** The blade is long compared with λ, so nothing varies with y. At MHz frequencies the free-space wavelength (hundreds of metres) dwarfs the electrodes, so the field obeys electrostatics.

(A2) **Periodic surface potential.** On $z = 0$ the potential $f(x)$ satisfies $f(x + \lambda) = f(x)$. Its values across the gaps between electrodes are never needed, only its period.

(A3) **Uniform soil.** The half-space $z > 0$ is linear, isotropic and homogeneous, with no free charge. For conductive soil at angular frequency ω, replace ε by $\varepsilon - j\sigma/\omega$; a homogeneous medium still obeys Laplace's equation, so Section 2 is unchanged.

(A4) **Bounded far field.** V stays bounded as $z \to \infty$.

2 The potential in the soil: the $\lambda/2\pi$ rule

With no free charge in a uniform medium, Gauss's law reduces to Laplace's equation:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial z^2} = 0, \qquad z > 0. \tag{1}$$

By (A2) the surface potential has a Fourier series,

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos k_n x + b_n \sin k_n x), \qquad k_n = \frac{2\pi n}{\lambda}. \tag{2}$$

Separation of variables. Try $V = X(x)\,Z(z)$. Substituting into (1) and dividing by XZ gives $X''/X = -Z''/Z$. The left side depends only on x and the right only on z, so both equal a constant, $-k^2$:

$$X'' + k^2 X = 0, \qquad Z'' - k^2 Z = 0. \tag{3}$$

Periodicity (A2) admits only $k = k_n$, with $X = \cos k_n x$ or $\sin k_n x$. Then $Z = A\,\mathrm{e}^{-k_n z} + B\,\mathrm{e}^{+k_n z}$, and for $n \geq 1$ the growing term violates (A4), so $B = 0$. For $n = 0$, $Z = A + Bz$, and (A4) again forces $B = 0$. Superposing, and matching (2) at $z = 0$:

$$\boxed{V(x,z) = a_0 + \sum_{n=1}^{\infty} (a_n \cos k_n x + b_n \sin k_n x)\,\mathrm{e}^{-2\pi n z/\lambda}} \tag{4}$$

This satisfies (1) term by term, equals f on $z = 0$, and is bounded. It is the only such solution.[1] The field $\mathbf{E} = -\nabla V$ is

$$E_x = \sum_{n=1}^{\infty} k_n (a_n \sin k_n x - b_n \cos k_n x)\,\mathrm{e}^{-k_n z}, \qquad E_z = \sum_{n=1}^{\infty} k_n (a_n \cos k_n x + b_n \sin k_n x)\,\mathrm{e}^{-k_n z}. \tag{5}$$

The constant a_0 carries no field. Harmonic n decays at rate $k_n = 2\pi n/\lambda$, so relative to the fundamental it is suppressed by $\mathrm{e}^{-(n-1)2\pi z/\lambda}$, and a short distance from the surface only $n = 1$ remains. Its 1/e depth is

$$\boxed{\delta = \frac{1}{k_1} = \frac{\lambda}{2\pi}} \tag{6}$$

Only λ appears. Finger count, finger length and board area do not, and by (A3) neither do frequency or conductivity. Electrode widths enter only through the coefficients a_n, b_n: they set how strong the field is, not how fast it decays.

3 Where the signal comes from: energy, not field

Capacitance measures stored energy, $W = \frac{1}{2} C V_1^2$. To first order, raising the permittivity by $\delta\varepsilon$ over a region Ω of the cross-section changes the capacitance per unit length by

$$\delta C = \frac{1}{V_1^2} \int_{\Omega} \delta\varepsilon\, |\mathbf{E}|^2 \,\mathrm{d}A, \tag{7}$$

where the field redistribution contributes only at second order. The share of the response from soil within depth d is therefore the share of $|\mathbf{E}|^2$ stored there, not the share of $|\mathbf{E}|$.

[1] The difference of two bounded solutions is harmonic and vanishes on $z = 0$. Its odd reflection into $z < 0$ is bounded and harmonic on the whole plane, hence constant by Liouville's theorem, hence zero.

For a single harmonic, (5) gives $E_x^2 + E_z^2 = k_n^2 c_n^2\, \mathrm{e}^{-2k_n z}$ at every x, with $c_n^2 = a_n^2 + b_n^2$, because $(a\sin\theta - b\cos\theta)^2 + (a\cos\theta + b\sin\theta)^2 = a^2 + b^2$. Cross terms between different harmonics average to zero over one period, so

$$\langle |\mathbf{E}|^2 \rangle_x(z) = \sum_{n=1}^{\infty} k_n^2 c_n^2\, \mathrm{e}^{-2k_n z}. \tag{8}$$

The exponent is $2k_n$: energy decays twice as fast as the field. Integrating from 0 to d and normalising,[2]

$$S(d) = \frac{\int_0^d \langle |\mathbf{E}|^2 \rangle_x \,\mathrm{d}z}{\int_0^\infty \langle |\mathbf{E}|^2 \rangle_x \,\mathrm{d}z} = \frac{\sum_n k_n c_n^2 (1 - \mathrm{e}^{-2k_n d})}{\sum_n k_n c_n^2}. \tag{9}$$

For the fundamental alone,

$$\boxed{S_1(d) = 1 - \mathrm{e}^{-4\pi d/\lambda}} \tag{10}$$

whose 1/e depth is $\lambda/4\pi$; half the signal comes from within $d_{1/2} = (\ln 2/4\pi)\,\lambda \approx 0.055\,\lambda$. Because $k_n \geq k_1$, every bracket in (9) is at least $1 - \mathrm{e}^{-2k_1 d}$, so

$$S(d) \;\geq\; 1 - \mathrm{e}^{-4\pi d/\lambda}. \tag{11}$$

Harmonics only make the response shallower; the fundamental sets the deepest reach.

Note. The rule $1 - \mathrm{e}^{-2\pi d/\lambda}$ in the 2026-09-15 architecture note uses the decay rate of the field in place of that of the energy. It matches neither limit derived here, though its initial slope, $2\pi d/\lambda$, equals that of the strong-contrast limit in Section 4.

4 Strong contrast: soil over air

Equation (7) is first order in $\delta\varepsilon$. The field solve instead fills $0 < z < d$ with soil (ε_1) and $z > d$ with air (ε_2), and reports

$$S(d) = \frac{C(d) - C(0)}{C(\infty) - C(0)}$$

at $\varepsilon_1/\varepsilon_2 = 20$, which is far from a small perturbation. For one harmonic it can be solved exactly. Hold the surface potential at $V_0 \cos kx$.[3] The solutions of (1) are

$$V = (A\,\mathrm{e}^{-kz} + B\,\mathrm{e}^{kz})\cos kx \quad (0 < z < d), \qquad V = C\,\mathrm{e}^{-kz}\cos kx \quad (z > d). \tag{12}$$

At $z = d$ the potential and the normal displacement εE_z are continuous:

$$A\,\mathrm{e}^{-kd} + B\,\mathrm{e}^{kd} = C\,\mathrm{e}^{-kd}, \qquad \varepsilon_1(A\,\mathrm{e}^{-kd} - B\,\mathrm{e}^{kd}) = \varepsilon_2\, C\,\mathrm{e}^{-kd}. \tag{13}$$

Eliminating C,

$$B = r\,\mathrm{e}^{-2kd} A, \qquad r = \frac{\varepsilon_1 - \varepsilon_2}{\varepsilon_1 + \varepsilon_2}. \tag{14}$$

The charge the soil draws from the electrode plane is $D_z(0^+) = \varepsilon_1 k(A - B)\cos kx$, with $A + B = V_0$. Writing it as $\varepsilon_{\mathrm{eff}}\, kV_0 \cos kx$ defines the permittivity the electrodes effectively see:

$$\varepsilon_{\mathrm{eff}}(d) = \varepsilon_1 \frac{1 - r\,\mathrm{e}^{-2kd}}{1 + r\,\mathrm{e}^{-2kd}} = \varepsilon_1 \frac{\varepsilon_2 + \varepsilon_1 \tanh kd}{\varepsilon_1 + \varepsilon_2 \tanh kd}. \tag{15}$$

[2] The sums converge because the field energy per period is finite; the $r^{-1/2}$ field singularity at an electrode edge is square-integrable.

[3] For a real multi-harmonic pattern this assumes the potential across the gaps does not redistribute as the soil changes. It is exact for a single harmonic.

The soil-side capacitance is proportional to ε_{eff}, with $\varepsilon_{\text{eff}}(0) = \varepsilon_2$ and $\varepsilon_{\text{eff}}(\infty) = \varepsilon_1$. With $\rho = \varepsilon_2/\varepsilon_1$ and $t = \tanh kd$,

$$\boxed{S(d) = \frac{\varepsilon_{\text{eff}} - \varepsilon_2}{\varepsilon_1 - \varepsilon_2} = \frac{(1+\rho)\,t}{1+\rho\,t}, \qquad t = \tanh\frac{2\pi d}{\lambda}} \tag{16}$$

Limits. As $\rho \to 1$ (weak contrast), $S = 2t/(1+t) = 1 - \mathrm{e}^{-2kd}$, recovering (10). As $\rho \to 0$ (strong contrast), $S = \tanh kd$. Since $\partial S/\partial\rho = t(1-t)/(1+\rho t)^2 \geq 0$, strong contrast gives the deepest response, and

$$\tanh\frac{2\pi d}{\lambda} \;\leq\; S(d) \;\leq\; 1 - \mathrm{e}^{-4\pi d/\lambda}. \tag{17}$$

Median depth. Setting $S = \frac{1}{2}$ gives $t = 1/(2+\rho)$, so

$$d_{1/2} = \frac{\lambda}{2\pi}\,\operatorname{artanh}\frac{1}{2+\rho}. \tag{18}$$

For soil at $\varepsilon_r = 20$ over air, $\rho = 0.05$ and $d_{1/2} = 0.0849\,\lambda$. Since S increases with kd, higher harmonics again only make the response shallower, so (18) with $k = k_1$ bounds the depth of a periodic pattern from above.

5 Comparison with the field solve

Electrode	λ	$\lambda/2\pi$ (1/e field)	field rate $\frac{\ln 2}{2\pi}\lambda$	energy, weak contrast (10)	soil over air $\rho = \frac{1}{20}$ (18)	2D field solve
MSV3 IDE (0.5 / 1.0)	3.0	0.48	0.33	0.17	0.25	≈0.23
v1.2 (G3.5 / 2 / H9.5 / 2 / G3.5)	17.0	2.71	1.88	0.94	1.44	1.39
MSV4 as drawn (A|B + guard)	18.9	3.01	2.09	1.04	1.60	1.01

Table 1: Median depth $d_{1/2}$ in mm ($\lambda/2\pi$ is the 1/e field depth, for reference). Solve: lossless soil $\varepsilon_r = 20$ over air, 1.6 mm FR4, 20 μm mask; medians interpolated linearly between solved depths. All values are modelled; none is measured.

- Equation (18) is the only one of these that uses the same definition as the solve. It is within 4% of the solve for the v1.2 and within about 10% for MSV3.
- MSV4 is shallower than (18) predicts (1.01 vs 1.60 mm). Its pattern is not a single period: the plate-to-guard field across the 1.04 mm clearance is a strong short-wavelength component, and by (9) short wavelengths are shallow. This is the guard-ring effect the 2026-09-19 field-solve note reports.
- A real blade is finite, so its surface pattern has a continuous spectrum that includes wavelengths longer than λ, which decay more slowly. This is the likely origin of the solve's deep tail: at 8 mm it gives 96.2% (v1.2) and 95.3% (MSV4), where (16) gives 99.5% and 99.1%.

Sources (repository `MSV4/`): field solve `scripts/msv4_electrode_fieldsolve.py`; v1.2 and MSV4 values `generated/electrode_compare_2026-09-23.json`; MSV3 values `docs/MSV4_ELECTRODE_FIELD_SOLVE_2026-09-19.md`; medians `scripts/msv4_sensing_depth_figure.py`.
Background: A. V. Mamishev *et al.*, "Interdigital sensors and transducers," *Proc. IEEE* 92(5), 808–845, 2004.